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Going Wireless In the Enterprise



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Wireless Enterprise Access Outline

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- Why Wireless?
- Challenges in Designing a Solution
- Wireless Enterprise Access Architecture
- Certicom Solutions
- Palm VII System

Mobile Enterprise Access
Why Now?

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- Increasing number of road-warriors
  - Access to e-mail
  - Access to corporate data
  - Access to Internet
- Need for improved productivity
- Competitive advantages
- Explosion in PDA, handheld market
- Enterprise managers going “hi-tech”

Mobile Enterprise Access Going Wireless

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- Real-time, on-demand access to data
- Wireless data is here…finally!
  - Networks are data ready
  - Increasing investment from service providers
- Handheld access to wireless made easy
  - Innovative user interface designs
  - Ready-to-go data capability
- Availability of strong, efficient security mechanisms over wireless networks

Designing a Mobile Enterprise Solution

- Various types of data networks
  - e.g., Cellular Packet Data, Wireless LANs, Wireline
  - Different transport and security protocols
- Various types of devices
  - Applications must be OS-agnostic
  - Varying level of “programmability”
- Synchronization of data across devices, systems
- Security considerations for the application

Challenges in Wireless Environment

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- Wireless medium makes attacks possible
  - Cloning
  - Eavesdropping
- Typically constrained devices
  - Limited storage
  - Limited computational resources
  - Battery life issues
- Bandwidth limitations over the air-link
- Air-link security vs. application-level security

Wireless Enterprise Access Security Objectives

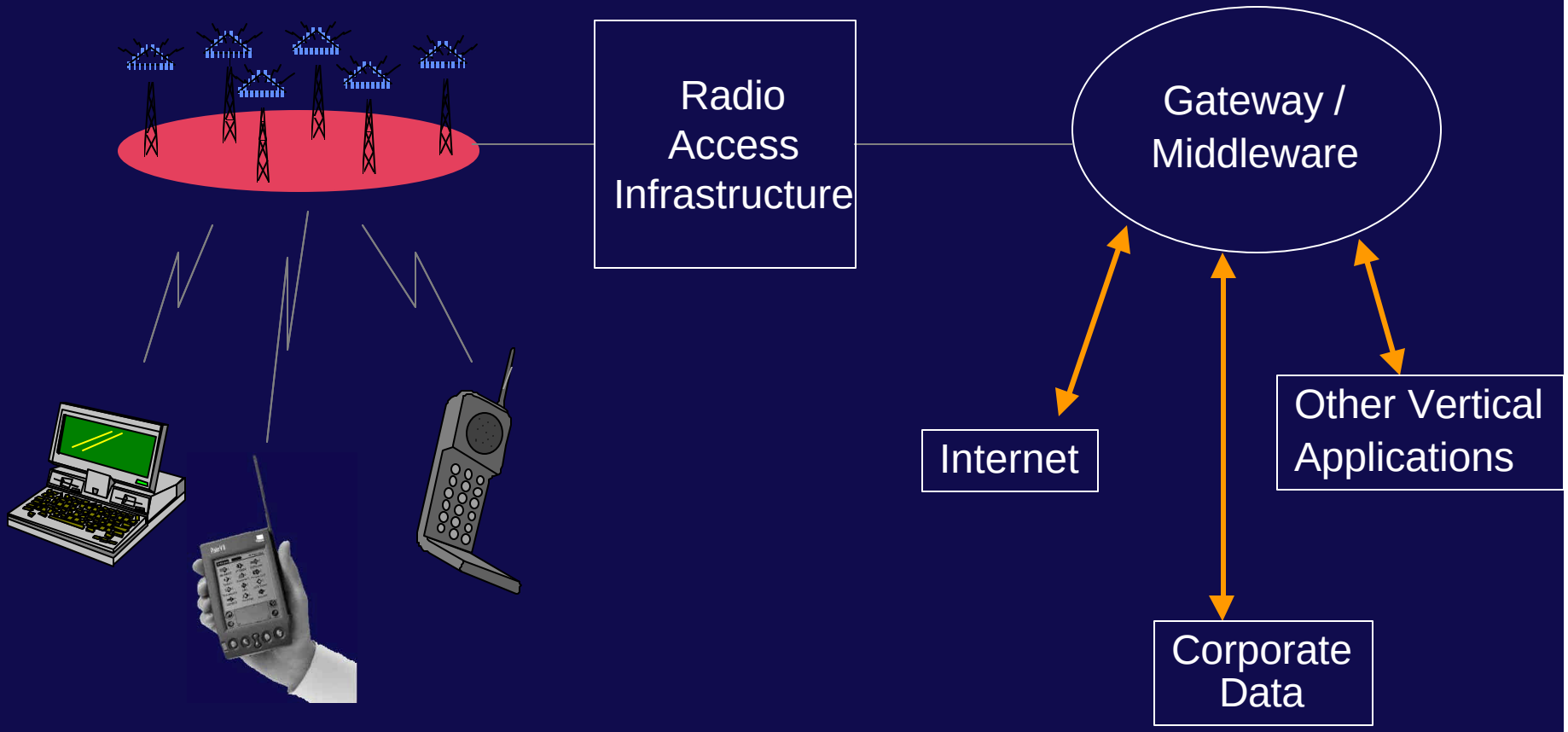
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- Meet the requirements of devices
- Efficient usage of bandwidth
- Secure application: authentication and privacy
- End-to-end security is desirable
- Ability to provide non-repudiation (e.g., financial transactions)

Wireless Enterprise Access Typical Solution

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Insertion of middleware or application gateway



Wireless Enterprise Access Middleware Solution

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- Gateway/Middleware provides:
  - Protocol translation (e.g., SMS - IP)
  - Content re-formatting (e.g., Web Clipping)
- Enables interoperability relieving client's burden
- One architecture --> many services, many devices
- Must also provide end-to-end security

Wireless Enterprise Access Security Lifecycle

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- Over-the-air service activation
- Session authentication and key agreement
  - Device authenticates the gateway
  - Device and gateway establish session keys
- Session communication
  - Message integrity
  - Privacy
- User authentication over the encrypted wireless link
- Back-end security (e.g., via SSL, private links)

Wireless Enterprise Access Adding Security

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Certicom protocol toolkits provide a complete solution

Wireless Link Security

Gateway or Server

Internet

Back-end Security

Corporate Data

- **SSL Plus for Embedded Systems**
- **WTLS Plus for WAP Security**
- **One Pass Protocol**

SSL Plus

Note: Gateway may reside behind the firewall for some applications—e.g., e-banking

Certicom Security Solutions

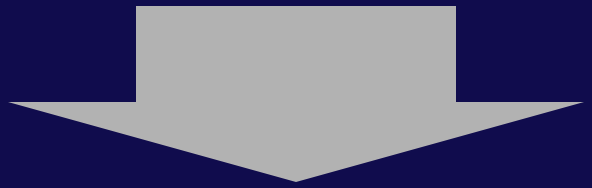
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- Communication security for applications and networks
  - Palm VII™ device with Palm.Net service
  - Middleware solutions for mobile enterprise access (AvantGo, Riverbed, JP Systems, Aether)
- Cryptographic libraries for defining secure applications using SDKs
  - BellSouth RAMpowertool™
  - JP Systems' One Touch Messaging Platform
- Security Builder SDK for adding security to applications

Certicom Toolkits

Protocol Solutions

SSL Plus	SSL Plus for Embedded Systems	SSL Plus Java	WTLS Plus	Customized Protocols (Palm VII)



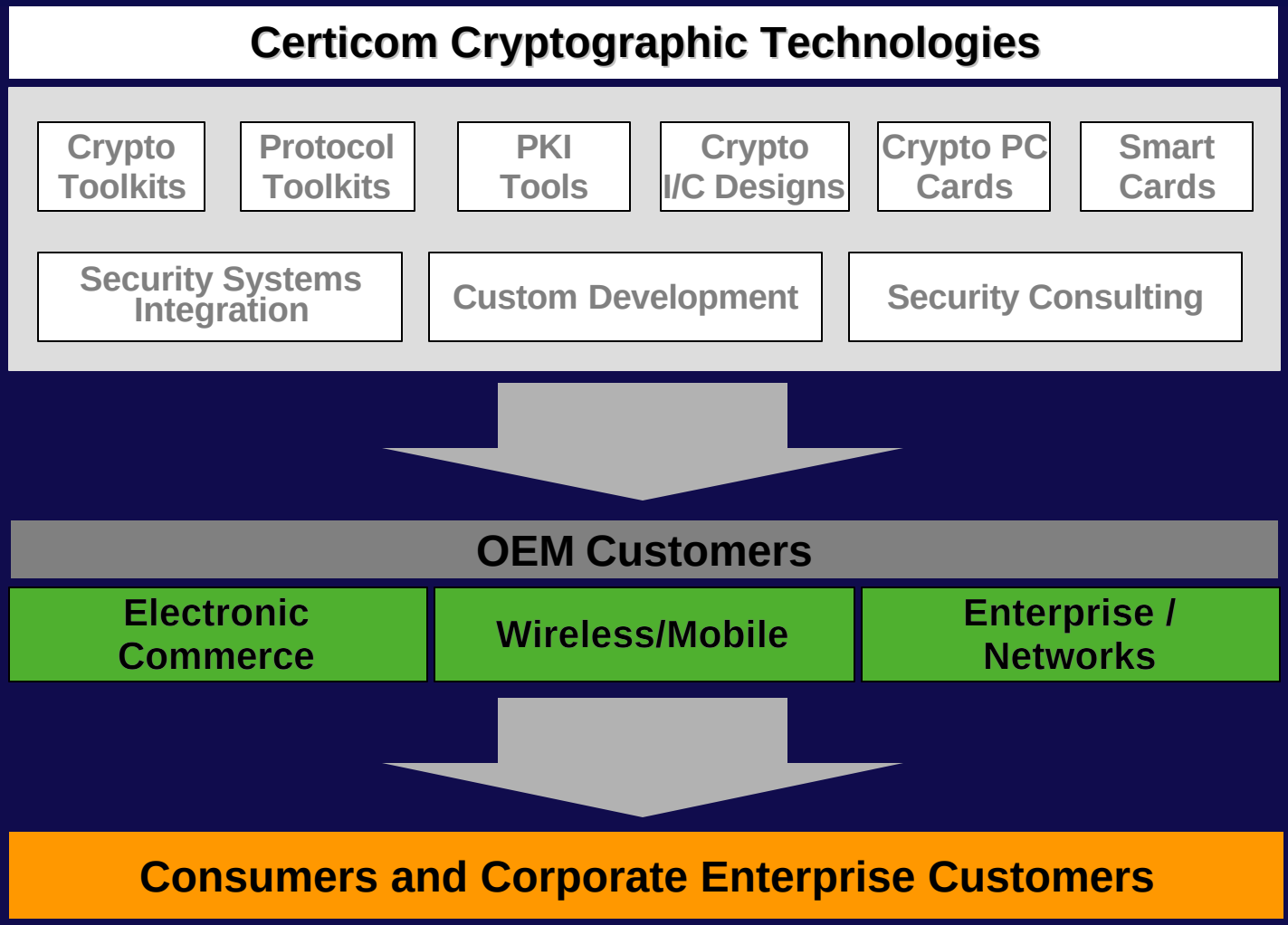
Security Builder

Core Cryptographic Toolkit

- Protocol solutions come packaged with Security Builder
- Security Builder can also be licensed separately

Certicom—
Full Service Security Provider

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Certicom Licensees and Partners Wireless Enterprise

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3Com/Palm Computing

724 Solutions

Actual Software

Aether Technologies

ASL

AvantGo

BellSouth

Infowave

JP Systems

Newcom Technologies

Puma Technologies

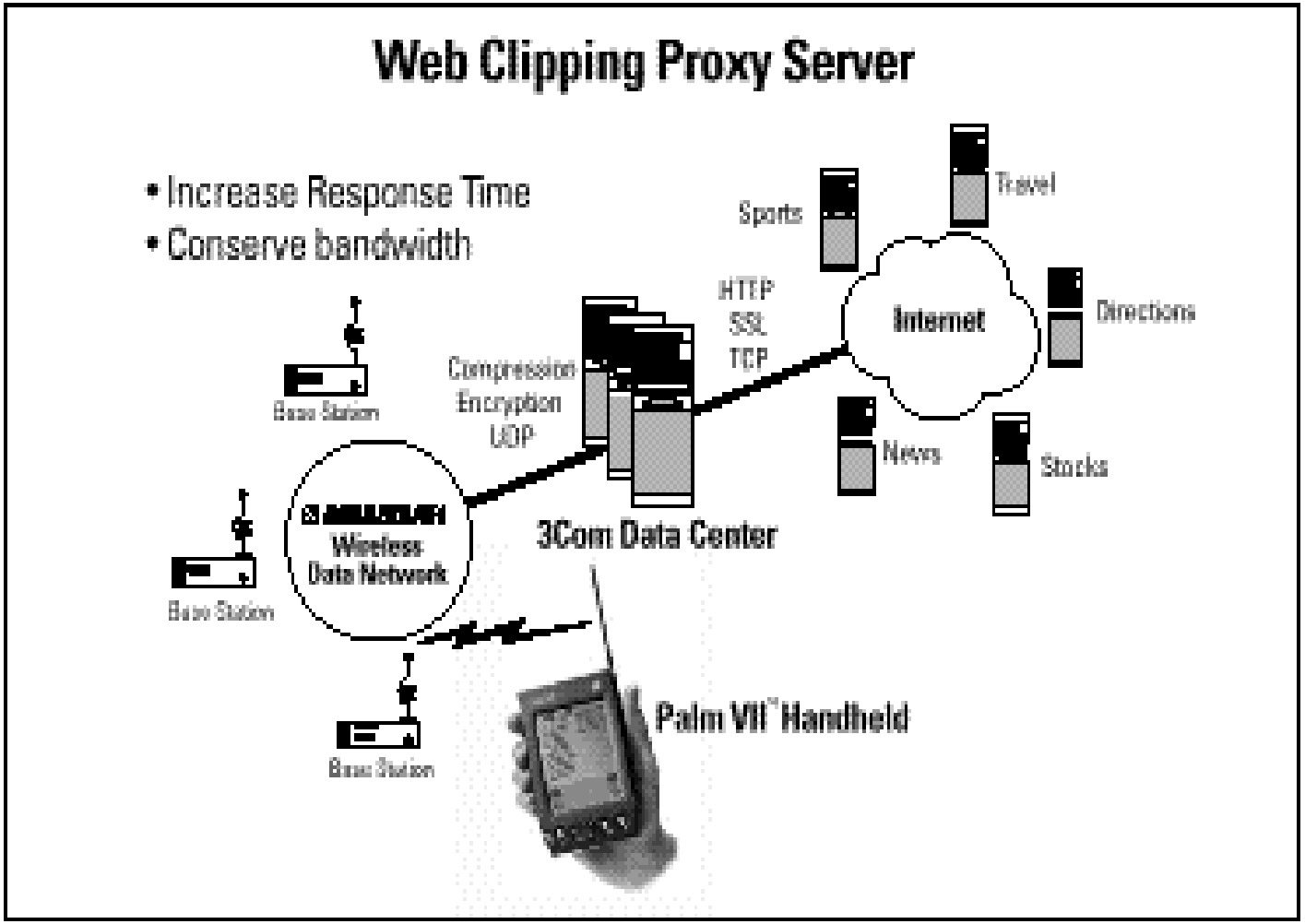
RIM

Riverbed Technologies

Schlumberger

Palm VII™ System

Palm VII™ System Architecture



Source: PalmVII whitepaper available from www.palm.com

Palm VII™ System Solution

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- 3Com provides the proxy server
- BellSouth provides the data network
- Certicom secures the wireless link
- Content providers can join Palm.Net and extend their service to mobile users
- Can be easily extended to other devices
- Vast number of enterprise and e-commerce applications

Certicom Branding

- Targeted at the end users
- Third-party trust
- Co-marketing opportunities
- Bringing together our partners
- Interoperability benefits the Palm Economy





Conclusion

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- Wireless enterprise/Internet access is booming
  - Palm platform is leading the market definition
  - Partnerships are key to a successful solution
- Security is a requirement—build it in!
- SDKs available to develop secure applications for the Palm Computing® platform
- Certicom provides industry-leading security solutions for the Palm platform
  - Network technology agnostic
  - Efficient cryptography and protocol solutions

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Going Wireless in the Enterprise



Greg Pinter
Vice President of Business Development

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Agenda

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- JP Systems Overview
  - Corporate Vision
  - Structure
  - Products using Wireless
- Enterprise Solutions/Case Studies
  - Pre-paid Phone Cards
  - KISS Products
- OTMP Wireless SDK
- Security/Encryption

JP Systems Mission

Provide innovative and easy to use software products and solutions for anywhere, anytime data access using wireless handheld terminals



JP Systems Vision— Simplifying Wireless Data

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Client Based Products

Wireless Devices: PDAs, Phones, Pagers

JP Systems

Server Based Products

Internet Content

- Remote Mail
- Scheduling
- Database
- Transactions

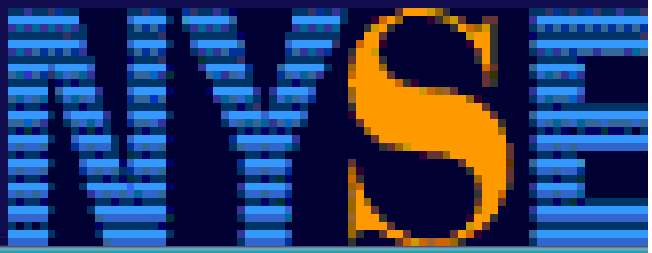
Corporate Data

JP Systems provides Server- and Client-based products to simplify wireless connectivity

Enterprise Solutions

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- Designed and developed mission-critical wireless advanced messaging system for NYSE
  - Busiest customer owned messaging system in the world
  - Highly reliable and robust server technology
  - Each message delivered end-to-end in less than 4 seconds
- Provides the critical communications between the brokers and their company's booth on the trading floor
- Touch screen messaging system
- JPS leverages this experience into all our other Server/Client products



Prepaid Phone Cards

- 2000 vending locations (prepaid phone cards) required cash pickup, deposit, and verification to corporate office
- No real time tracking of pickup and deposits
  - Delays caused uncertainty of cash collected and deposited
- Inaccuracies in recorded field data vs. office input data (SQL database)
- Extensive manpower needed to receive faxed data and enter into database

Prepaid Phone Cards

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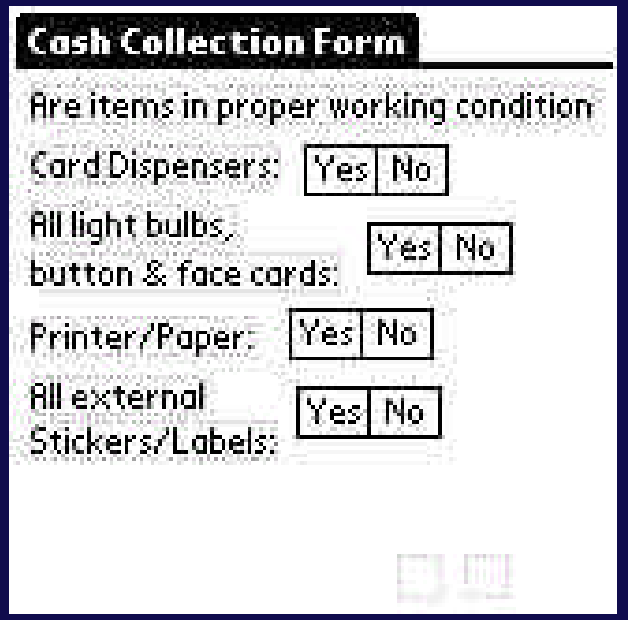
- JP Systems and Palm Computing® solution:
  - JPS custom client application on Symbol SPT 1500 with Glenayre AccessLink II device
  - JPS server software receives data instantly as e-mail, parses into database, and sends receipt
  - Wireless end-to-end solution gets pick up and deposit information to corp. office immediately, accurately with no human intervention

Prepaid Phone Cards

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Kiss Products

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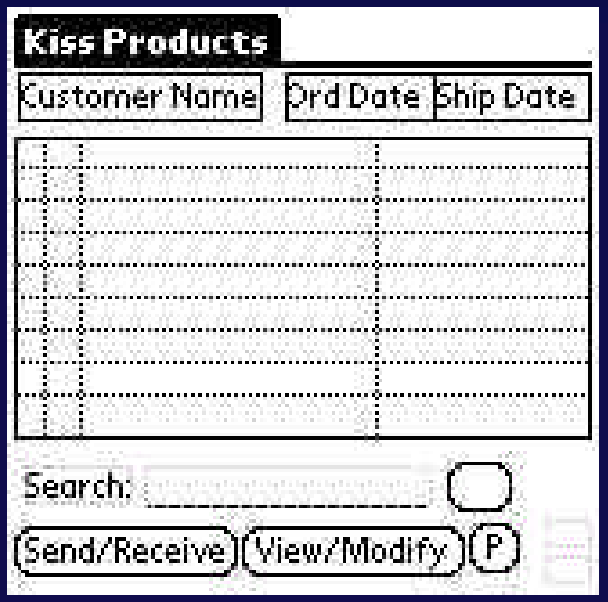
- NY based cosmetic distributor with field sales
- Needs updated inventory, pricing and customer records on site
- Large cumbersome “book” of information needed constant updating; long look up time
- Orders needed to be manually recorded and faxed for processing

Kiss Products

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- JP Systems/Palm Computing® solution:
  - Customer records and updated inventory pulled daily with Palm modem or GSM phone
  - Fast search and look up tool created on client application to quickly get to the item needed
  - Customer Orders saved in outbox to send when convenient
  - Orders input into AS400 database for processing

Kiss Products







One Touch Messaging Platform

An SDK that allows developers to easily and quickly incorporate wireless communications into their Palm Application

- Componentized program for custom development
- Licensing model for developers
- Private labeling available
- Module-aware for transparent user experience (pagers, modems, cell phones all work out of same program)

OT Message Platform

Wireless Connectivity for Palm OS®

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Novatel Minstrel



Glenayre
AccessLink II



Ericsson
i888





Palm,
Workpad, and
Handspring
Modems



Outlook Synchronization

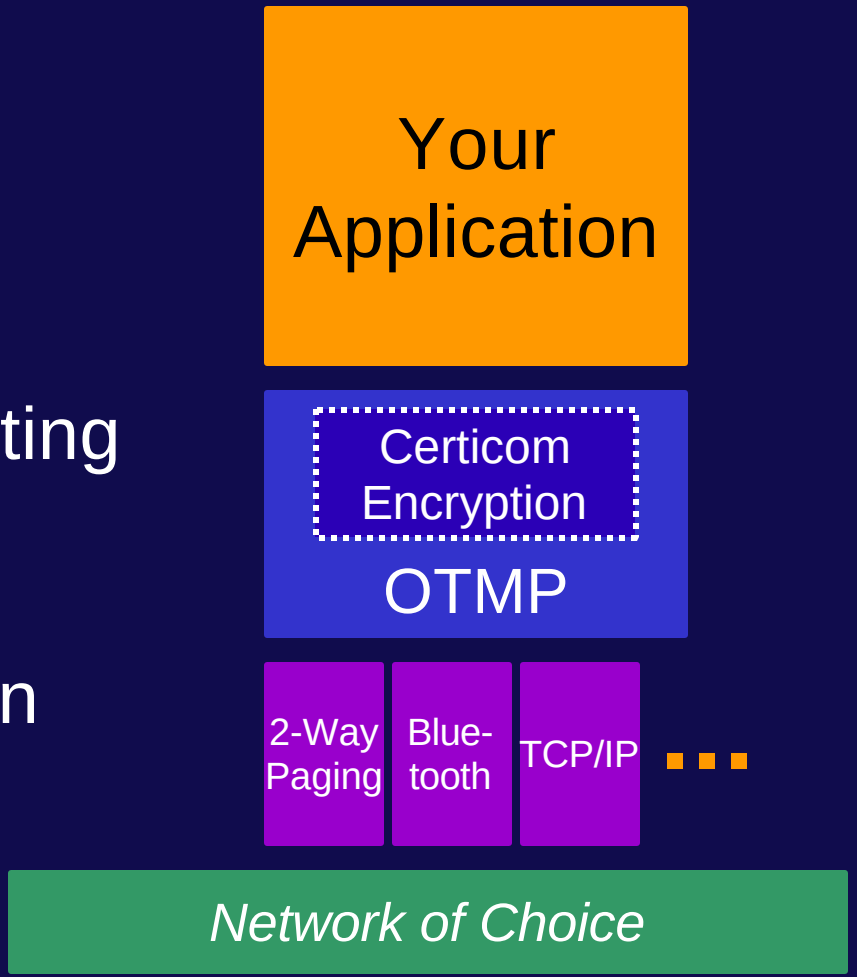


Nokia 7100

OT Message Platform

Wireless SDK Concept

- Advanced Message Architecture
- Provides a Unified, Consistent Interface
- Plug-and-play with existing applications
- Optional Certicom's Elliptic Curve Encryption

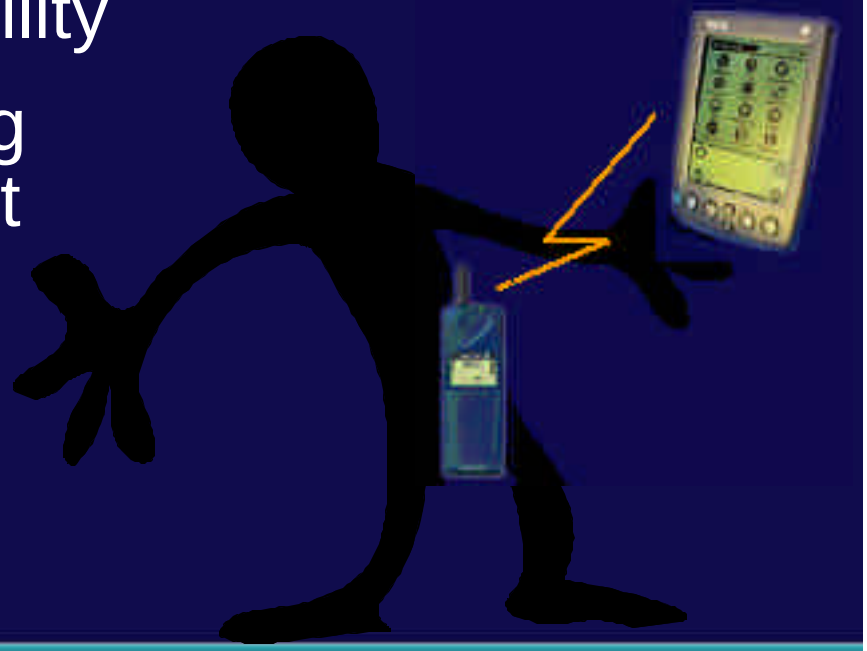


OT Message Platform

Bluetooth Is Coming!

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- Bluetooth enabled devices will provide the glue for connecting “companions”
- Users won’t be tied to small screens to read WAP information
- Convenience and portability
- JP Systems is supporting Widcomm’s BlueConnect



Available Plug-In Modules

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- POP3/SMTP
- IMAP4
- 2-Way Paging
- CDPD
- GSM
- DSP
- Bluetooth

OTMP Licensees

One-Touch Mail and InfoBeam

- OpenSky
- PocketScience
- GoAmerica
- Palm Asia
- Glenayre
- Motorola
- WidComm
- Barpoint

Security and Encryption

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- JP Systems has licensed from Certicom:
  - SSL Plus
  - Security Builder
- Enables encryption to be included in all wireless connections for the Palm
  - High strength Elliptic Curve Cryptography
  - Compact and airtime efficient
- Allows secure communications for transactions and corporate data

Summary

- JP Systems is the leader in technology to wirelessly enable the Palm™ family of products
- Using our toolset, seamless connectivity with all networks and protocols
- OTMP provides the network agnostic wireless solution to application developers
- Contact us at www.jpsystems.com


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